

PLANT KINGDOM

1. Which group does not produce embryo ?
 (1) Algae (2) Moss
 (3) Liverworts (4) Club moss
2. Which group of plants conduct water and minerals by xylem tracheids ?
 (1) Algae (2) Mosses
 (3) Liverworts (4) Gymnosperms
3. In which group the male and female sex organs are called antheridia and archegonia, respectively ?
 (1) Blue green algae (2) Eubacteria
 (3) Protista (4) Bryophyta
4. The group of plants, in which body is differentiated into root, stem and leaf :-
 (1) Chlorophyceae (2) Phaeophyceae
 (3) Lycopsidea (4) Hepaticopsida
5. Which is not an example of moss plants ?
 (1) *Funaria* (2) *Polytrichum*
 (3) *Sphagnum* (4) *Colletotrichum*
6. Strobilli or cones are not formed in :-
 (1) Lycopods (2) Sphenopsids
 (3) Conifers (4) Ferns
7. In which group of plants both male and female gametophytes do not have an independent free living existence ?
 (1) Bryophytes (2) Pteridophytes
 (3) Gymnosperms (4) Both 1 and 2
8. Sporophylls are arranged spirally along an axis, when they aggregate, to form :-
 (1) Strobillus (2) Flowers
 (3) Inflorescence (4) Thalamus
9. The gametophyte of pteridophytes require to grow :-
 (1) Warm, damp, and shady place
 (2) Cool, damp, and shady place
 (3) Warm, dry, and shady place
 (4) Cool, dry, and place of well sunshine
10. Zygotic meiosis occurs in :-
 (1) *Pinus* (2) *Funaria*
 (3) *Pteridium* (4) *Chara*
11. The megaspore mother cell in gymnosperms is differentiated from :-
 (1) Integument (2) Embryosac
 (3) Nucellus (4) Endosperm
12. Which is not common in *Chlorella* and *Spirulina* ?
 (1) both are unicellular
 (2) both are rich in protein
 (3) both are used as food supplement
 (4) both are prokaryotes
13. Majority of red algae tend to grow in :-
 (1) Marine and warmer areas
 (2) Marine and colder areas
 (3) Freshwater and warmer areas
 (4) Brackish water and colder areas
14. Pyriform, biflagellate gametes are produced in:-
 (1) *Ectocarpus*, *Polysiphonia*
 (2) *Ectocarpus*, *Laminaria*
 (3) *Ulothrix*, *Polysiphonia*
 (4) *Fucus*, *Porphyra*
15. Autotrophic aquatic organisms which usually reproduce vegetatively by fragmentation, asexual by non motile spores and perform sexual reproduction also by the non-motile gametes. These organisms are :-
 (1) *Polysiphonia*, *Porphyra*, *Gracilaria*
 (2) *Ectocarpus*, *Dictyota*, *Laminaria*
 (3) *Laminaria*, *Fucus*, *Sargassum*
 (4) *Volvox*, *Chara*, *Spirogyra*
16. Which is not true about agar ?
 (1) It is obtained from red alga
 (2) It is used to grow (culture) microbes
 (3) It is used to make ice-cream and gellies
 (4) It is used as food supplement even by space travellers
17. At present most acceptable system of classification is :-
 (1) Artificial system (2) Natural system
 (3) Phylogenetic system (4) Practical system

18. Fill in the blanks a, b, c and d by observing the characters given in the table and choose the correct answer from the following options :-

Plant group	Main body	Fertilisation	Vascular tissue	Female sex organ
Bryophyta	Gameto-phyte	By Zoido-gamy	Absent	(c)
Pterido-phyta	(a)	By Zoido-gamy	(b)	Arche-gonium
Gymno-sperm	Sporo-phyte	By Siphono-gamy and Zoidogamy	present	(d)

(a) (b) (c) (d)

- (1) Sporophyte Present Archegonium Archegonium
 (2) Sporophyte Absent Oogonium Archegonium
 (3) Gametophyte Present Oogonium Carpel
 (4) Gametophyte Present Archegonium Carpel
19. A group of algae, having following characters:-
 (a) Chlorophyll a and b are present
 (b) Chloroplast has one or many pyrenoids
 (c) Cell wall made of cellulose and pectose
 (d) Reserve food material is starch
 Which one of the following is also a very important character including above characters to call it as a member of chlorophyceae ?
 (1) Plant body colonial only
 (2) Plant body is made of haploid cells called as gametophyte
 (3) Its flagellar character may be 2-8 in number, equal and apical
 (4) It may occur in fresh water, brackish water or salt water
20. In which member chlorophyll a, c and fucoxantin are present ?
 (1) *Ectocarpus* (2) *Gelidium*
 (3) *Volvox* (4) *Vaucheria*
21. Floridean starch is characteristic feature of :-
 (1) *Polysiphonia*, *Gracilaria*, *Porphyra*
 (2) *Laminaria*, *Sargassum*, *Porphyra*
 (3) *Polysiphonia*, *Laminaria*, *Porphyra*
 (4) *Chara*, *Dictyota*, *Polysiphonia*
22. Gemmae are the structures of :-
 (1) Asexual reproduction produced in *Marchantia*
 (2) Sexual reproduction produced in most mosses
 (3) Asexual reproduction produced in most mosses
 (4) Sexual reproduction produced in most liverworts

23. Predominant stage of them is gametophyte, they vegetatively reproduce by fragmentation and budding. They produce male and female gametes in antheridia and archegonia, respectively. After fertilisation zygote develops into a sporophyte consisting of foot, seta and capsule, in them spore formed in their capsule form protonema on germination. The above description is about the group :-

(1) Lycopsidea (2) Bryopsida
 (3) Hepaticopsida (4) Psilopsida

24. Integumented megasporangia are formed in :-
 (1) Phanerogames (2) Cryptogames only
 (3) Gymnosperms only (4) Pteridophytes only
25. Evolutionary first terrestrial plants to possess vascular tissues xylem and phloem are :-
 (1) Psilopsids (2) Lycopods
 (3) Pteropsids (4) Sphenopsids
26. Needle like leaves to reduce the surface area, thick cuticle, and sunken stomata to reduce water loss etc. are the xerophytic characters present in :-
 (1) Pteridophytes
 (2) Gymnosperms
 (3) Angiosperms
 (4) Sphenopsida
27. Most reduced gametophyte is found in :-
 (1) Bryophyta
 (2) Pteridophyta
 (3) Gymnosperms
 (4) Angiosperms
28. Consider the following characters :-
 (A) Formation of only one functional megaspore in a megasporangium
 (B) Formation of hard covering around megasporangium
 (C) Development of embryo from zygote within the female gametophyte
 (D) Retention of megaspore inside the megasporangium
 These were very essential events occurred during the course of evolution for the phenomenon of:-
 (1) Heterosporous
 (2) Seed habit
 (3) Fruit formation
 (4) Covered seed formation

- 29.** In gymnosperms, the megaspore forms :-
(1) Endosperm (2) Female gamete
(3) Male gamete (4) Male cone
- 30.** Find the odd one with respect to ploidy?
(1) Spore (2) Sporophyte
(3) Archegonium (4) Antheridium
- 31.** Which of the following is odd one from the following about liverworts ?
(1) Foot, Seta, Capsule
(2) Spore, Archegonium, Antheridium
(3) Multicellular rhizoids, Primary protonema & Secondary protonema
(4) Rhizoids, Gemmae, Scales
- 32.** Which is not true about fern plants ?
(1) They produce sori on sporophylls
(2) They produce male and female cone by spiral arrangement of sporophylls
(3) Some fern plants are heterosporous also
(4) Some ferns are aquatic also
- 33.** In a sphenopsid plant number of chromosome in the cells of scaly leaves is 24, what will be the number of chromosomes in its cells of stem, spore, and prothallus respectively :-
(1) 24, 12, 12 (2) 24, 24, 12
(3) 12, 24, 12 (4) 24, 12, 24
- 34.** Which group of plants is of less economic importance but have ecological importance?
(1) Gymnosperms (2) Angiosperms
(3) Bryophyta (4) Pteridophyta
- 35.** Normally in gymnosperms how many archegonia are produced in an ovule ?
(1) One
(2) Two
(3) Two or more
(4) Always one
- 36.** Select the false statement ?
(1) Cyanobacteria, that are also referred to as blue-green algae are not algae any more
(2) Green algae, brown algae, red algae, bryophytes, pteridophytes, gymnosperms and angiosperms are included under the kingdom plantae
(3) Artificial systems gave equal weightage to vegetative and sexual characteristics.
(4) Cytotaxonomy uses the chemical constituents of the plant cells to resolve confusions of taxonomy.
- 37.** The habitat where algae can be found in ?
(1) Moist stones (2) Moist wood
(3) Moist soil (4) All of the above
- 38.** The most common type of spores produced in algae during asexual reproduction are :-
(1) Aplanospores (2) Conidia
(3) Zoospores (4) Hypnospores
- 39.** Fusion between a large, nonmotile (static) female gamete and a smaller, motile male gamete is termed as :-
(1) Isogamy
(2) Oogamy
(3) Anisogamy
(4) Autogamy
- 40.** Non vascular archegoniate plants are :-
(1) Thallophytes
(2) Bryophytes
(3) Pteridophytes
(4) Gymnosperm
- 41.** Vascular plants producing integumented megasporangia but not ovary are :-
(1) Pteridophytes
(2) Gymnosperms
(3) Angiosperms
(4) Bryophytes
- 42.** The same type of relation occurring in megaspore mother cell and nucellus can also be seen in :-
(1) Female gamete and antheridium
(2) Male gamete and endosperm
(3) Egg cell and archegonium
(4) Neck canal cell and antheridium
- 43.** Which is not true about the uses of algae ?
(1) They increase the level of dissolved oxygen in their immediate environment
(2) They are of paramount importance as primary producer of energy
(3) *Porphyra*, *Laminaria* and *Sargassum* are used as food
(4) Certain marine brown algae produce agar which is used to make ice-creams

- 44.** Which one of the following is not the ecological importance of moss plants ?
- (1) Some mosses provide food for herbaceous mammals birds and other animals
 - (2) Very high water holding capacity of mosses is useful for trans-shipment of living materials
 - (3) Mosses along with lichens are the pioneering organisms to colonise rocks
 - (4) Mosses form dense mats on the soil and reduce the impact of falling rain.
- 45.** In most of the plants reduction division occurs during the formation of :-
- (1) Male gamete and female gamete
 - (2) Male sex organ and female sex organ
 - (3) Microspore and megaspore
 - (4) Microsporangium and Megasporangium